

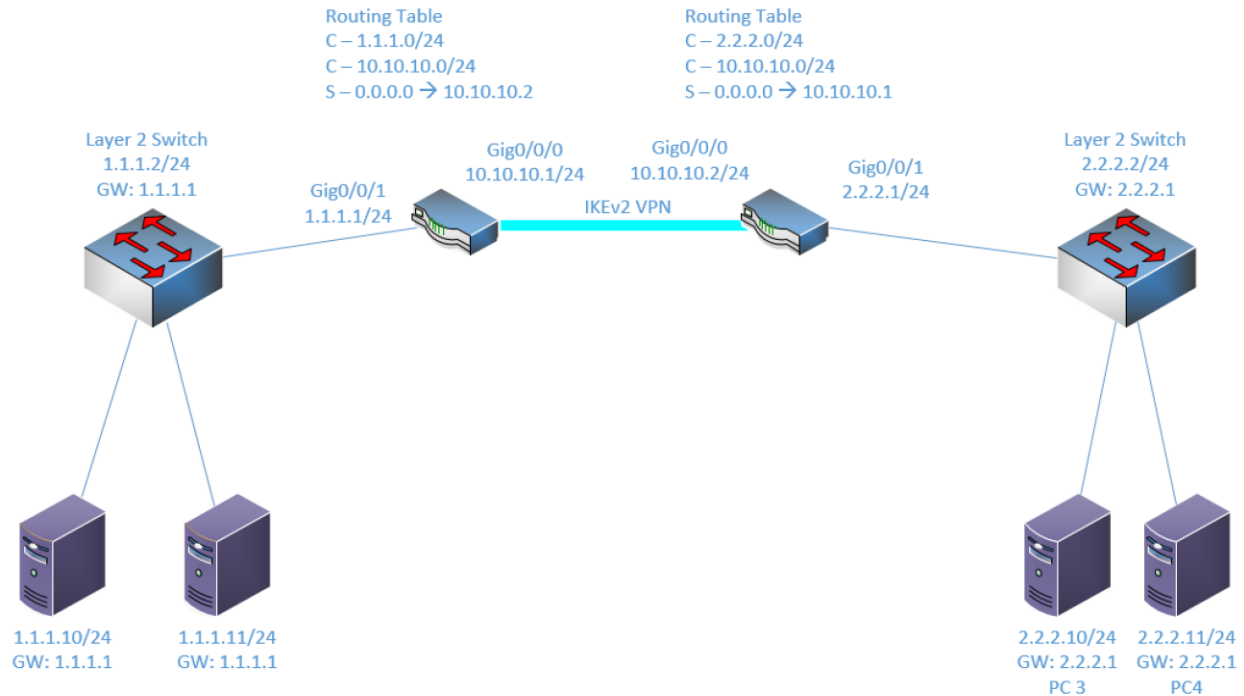
Lab 8

Implementing a IKEv2 VPN on IOS

Objective:

The purpose of this lab is for the student to understand configuration and implement an IKEv2 VPN. Students will utilize the IOS CLI to explore and configure an ISR Router.

Diagram:



Components:

- 4 – PCs
- 2 – Cisco 3650 Layer 2 Switches
- 2 – Cisco 4321 ISR Routers

Useful commands for this lab

```
show crypto ikev2 session !To display the session details
show crypto ikev2 sa detail !To display phase 1 details
show crypto ikev2 profile !To display the IKEv2 profile information
show crypto ipsec sa !To display the IPsec phase 2 details
```

Procedure:

1. Connect all PC's together on the layer 2 switches and put their static IP addresses, subnet masks and gateways on them. Test connectivity by pinging each machine within the network.
2. Configure the IP addresses on the Router's interfaces and turn on the interfaces.
3. Make sure the PCs on the same network can ping each other and their gateway.
4. Configure the static default route on each router as shown in the diagram
5. Use the supplied configuration to configure Router 1. configure each section separately so you can verify it's successful before moving on to the next
6. Use the configuration from Router 1 to create the configuration for Router 2. You'll have to remember to change IP addresses and names to be relevant to router 2. Technically the names could be the same, but it's best practice to change them. For Instance:
 - a. ROUTER1-PROFILE to ROUTER2-PROFILE
 - b. ROUTER1-ROUTER2-NACL to ROUTER2-ROUTER1-NACL
7. Before you add the configuration to router 2, make sure you change the name and any reference to the name lower in the config. For instance:
 - a. match address ROUTER1-ROUTER2-NACL to match address ROUTER2-ROUTER1-NACL

8. Once your text file is ready for Router 2, paste it in one section at a time so you can confirm that it's successful. If you have any typos in your names or references to names, you might get an error.
9. Once the crypto map is applied on the interfaces on both sides, you should be able to ping from 1.1.1.10 to 2.2.2.10. If not, verify your config and adjust until the configuration is successful.
10. Screen shot the VPN details from one of the routers and paste them below. Use the "show" commands provided above to verify your traffic is being encrypted.

```
NETC230-R2#show crypto ikev2 session
IPv4 Crypto IKEv2 Session

Session-id:1, Status:UP-ACTIVE, IKE count:1, CHILD count:1

Tunnel-id Local Remote fvrf/ivrf Statu
s
1 10.10.10.2/500 10.10.10.1/500 none/none READY

Encr: AES-CBC, keysize: 256, PRF: SHA512, Hash: SHA512, DH Grp:24, Auth si
gn: PSK, Auth verify: PSK
Life/Active Time: 86400/90 sec
Child sa: local selector 2.2.2.0/0 - 2.2.2.255/65535
remote selector 1.1.1.0/0 - 1.1.1.255/65535
ESP spi in/out: 0xCD007AB1/0x78225C57

IPv6 Crypto IKEv2 Session

NETC230-R2#
```

```

NETC230-R2#show crypto ikev2 sa detail
IPv4 Crypto IKEv2 SA

Tunnel-id Local Remote fvrf/ivrf Status
1 10.10.10.2/500 10.10.10.1/500 none/none READY
Encr: AES-CBC, keysize: 256, PRF: SHA512, Hash: SHA512, DH Grp:24, Auth sign: PSK, Auth verify:
PSK
Life/Active Time: 86400/126 sec
CE id: 1016, Session-id: 1
Status Description: Negotiation done
Local spi: AESF813B7E2692FD Remote spi: CDF6A1131F69878B
Local id: 10.10.10.2
Remote id: 10.10.10.1
Local req msg id: 2 Remote req msg id: 0
Local next msg id: 2 Remote next msg id: 0
Local req queued: 2 Remote req queued: 0
Local window: 5 Remote window: 5
DPD configured for 0 seconds, retry 0
Fragmentation not configured.
Dynamic Route Update: enabled
Extended Authentication not configured.
NAT-T is not detected
Cisco Trust Security SGT is disabled
Initiator of SA : Yes

IPv6 Crypto IKEv2 SA

NETC230-R2#
NETC230-R2#show crypto ikev2 profile

IKEv2 profile: ROUTER2-PROFILE
Ref Count: 2
Match criteria:
Fvrf: global
Local address/interface: none
Identities:
address 10.10.10.2 255.255.255.255
address 10.10.10.1 255.255.255.255
Certificate maps: none
Local identity: none
Remote identity: none
Local authentication method: pre-share
Remote authentication method(s): pre-share
EAP options: none
Keyring: KR-1
Trustpoint(s): none
Lifetime: 86400 seconds
DPD: disabled
NAT-keepalive: disabled
Ivrf: none
Virtual-template: none
mode auto: none
AAA AnyConnect EAP authentication mlist: none
AAA EAP authentication mlist: none
AAA Accounting: none
AAA group authorization: none
AAA user authorization: none
NETC230-R2#

```

```
NETC230-R2#show crypto ipsec sa

interface: GigabitEthernet0/0/0
  Crypto map tag: CMAP-ROUTER2, local addr 10.10.10.2

protected vrf: (none)
local ident (addr/mask/prot/port): (1.1.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (2.2.2.0/255.255.255.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
  #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

  local crypto endpt.: 10.10.10.2, remote crypto endpt.: 10.10.10.1
  plaintext mtu 1500, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0/0
  current outbound spi: 0x0(0)
  PFS (Y/N): N, DH group: none

inbound esp sas:

inbound ah sas:

inbound pcp sas:

outbound esp sas:

outbound ah sas:

outbound pcp sas:

protected vrf: (none)
local ident (addr/mask/prot/port): (2.2.2.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (1.1.1.0/255.255.255.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 15, #pkts encrypt: 15, #pkts digest: 15
  #pkts decaps: 15, #pkts decrypt: 15, #pkts verify: 15
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

  local crypto endpt.: 10.10.10.2, remote crypto endpt.: 10.10.10.1
  plaintext mtu 1422, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0/0
  current outbound spi: 0x78225C57(2015517783)
  PFS (Y/N): N, DH group: none
```

```

inbound esp sas:
 spi: 0xCD007AB1(3439360689)
 transform: esp-aes esp-sha512-hmac ,
 in use settings =(Tunnel, )
 conn id: 2002, flow_id: ESG:2, sibling_flags FFFFFFFF80000048, crypto map: CMAP-ROUTER2
 sa timing: remaining key lifetime (k/sec): (4607998/3423)
 IV size: 16 bytes
 replay detection support: Y
 Status: ACTIVE(ACTIVE)

inbound ah sas:

inbound pcp sas:

outbound esp sas:
 spi: 0x78225C57(2015517783)
 transform: esp-aes esp-sha512-hmac ,
 in use settings =(Tunnel, )
 conn id: 2001, flow_id: ESG:1, sibling_flags FFFFFFFF80000048, crypto map: CMAP-ROUTER2
 sa timing: remaining key lifetime (k/sec): (4607999/3423)
 IV size: 16 bytes
 replay detection support: Y
 Status: ACTIVE(ACTIVE)

outbound ah sas:

outbound pcp sas:
NETC230-R2#

```

11. Draw the lab diagram below (PC's, router and switch connections) Label the IP addresses, subnet masks, and gateways used by each device. Label the routing tables next to the routers. Label your IKEv2 VPN endpoints.

